

IntentPrism: Human-AI Intent Manifestation for Web Information Foraging

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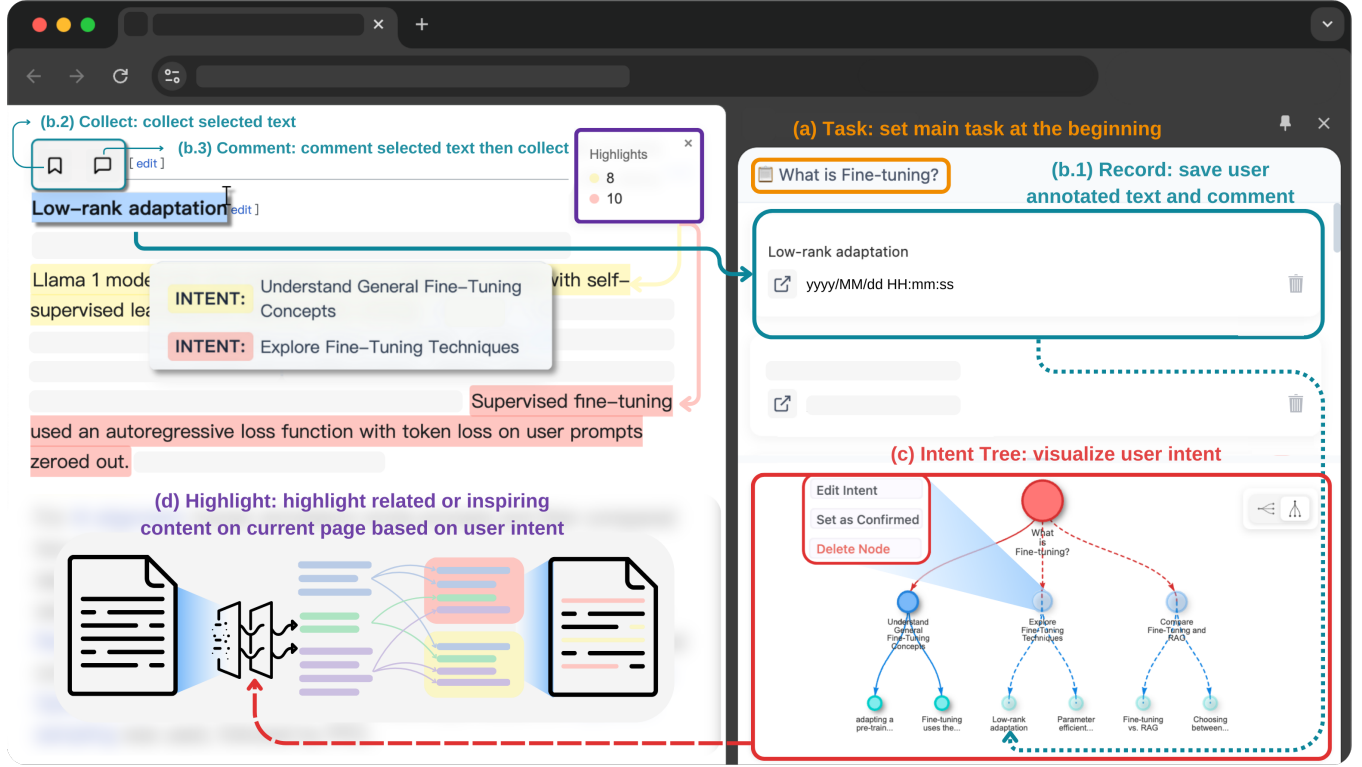


Figure 1: The interface of IntentPrism and illustration of functionality. This plug-in system activates a side panel (right) within the browser and integrates four mechanisms: (a) Task mechanism: users input their main goal of an information foraging task; (b) Recording mechanism: users save snippets of information by selecting and annotating pieces of content while browsing the internet; (c) Intent Tree mechanism: a dynamically constructed intent tree visualizing the relationships between levels of extracted user intentions and user-saved data snippets; (d) Highlighting mechanism: the system automatically highlights related content in yellow and potentially interesting new subjects in red based on intent analysis.

Abstract

Intent-based User Interface (IBI) is an emerging topic in HCI about supporting user interaction with intelligent systems. We introduce IntentPrism, an approach to explicitly visualizing LLM-recognized user intents in a web information foraging task, in order to facilitate human-AI collaboration. Implemented as a web browser plug-in, IntentPrism provides real-time intent analysis from bookmarked web content, intent tree visualization and relevant content highlighting on webpages. It aims to support users in refining vague intentions, improving focus, and uncovering relevant content. The preliminary user study ($n = 5$) showed the promising potential of this approach in refining coarse-grained intents and improving efficiency especially when conducting in-depth exploration tasks. Additionally, user feedback highlights the demands for a multi-layered intent structure and a feedback mechanism for inaccurate highlights. Overall, IntentPrism represents a novel exploration of designing advanced intent-based user interfaces.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**.

Keywords

Intent-based user interfaces, Large language models, Information foraging, User experience

ACM Reference Format:

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1 Introduction

Traditional user interfaces often required users to manipulate graphical elements or input specific commands. However, the rise of large language models (LLMs) has brought about a paradigm shift in interaction, enabling users to express their intent through natural language. This allows for a more flexible approach for users to express intent, bypassing the limitations of structured interfaces and predefined actions. In light of this recent development, there is an emerging interaction paradigm known as Intent-based User Interfaces (IUIs) [6]. Prior research has identified challenges users faced in formulating clear and actionable intentions (intent formation challenge) [23], articulating those intentions without ambiguity (intent expression challenge) [4, 19, 27], ensuring the system accurately interprets their input (intent recognition challenge) [2, 11, 15],

and confirming alignment between their intentions and the system's understanding (intent alignment challenge) [8, 9, 16, 17, 26].

Information Foraging theory applies animal foraging principles to explain how humans search for information [18]. It assumes users balance the expected value of information against its acquisition cost. With the emergence of LLM-driven tools such as Perplexity¹, the ways users perform information foraging tasks are undergoing significant transformation [5]. The powerful natural language processing capabilities of LLMs can accelerate information consumption. For instance, Perplexity uses techniques such as Chain-of-Thought (CoT) reasoning to decompose search intents, retrieve information, and generate summaries. However, users cannot specify sources, risking low-quality data inclusion. Tools like Elmo² addresses this by allowing users to operate within selected web pages via a side panel, providing control over sources. Yet, these tools rely on implicit intent decomposition, limiting precise intent alignment.

This work explores how to leverage AI models to dynamically analyse and support users' personalized intents in a web information foraging task. We aim to facilitate users' intention-driven knowledge work by building upon LLM-assisted dynamic intent analysis and extraction of relevant content. To reach this goal, this paper presents the first iteration of an intent-based web information foraging tool - IntentPrism, developed as a web browser plug-in. As introduced in Fig. 1, it supports four mechanisms: (a) Task mechanism enables users to establish research context by setting main goals, serving as root nodes for subsequent intent analysis; (b) Recording mechanism streamlines web content collection through a lightweight selection interface, enabling quick information capture and annotation; (c) Intent Tree mechanism visualizes hierarchical relationships between user intentions, distinguishing confirmed nodes (solid lines) from pending ones (dashed lines) while supporting manual modification; (d) Highlighting mechanism assists web browsing by automatically identifying both related contents (yellow highlights) and potentially inspiring new subjects (red highlights) based on semantic analysis of the current intent structure. We hope to iteratively develop this system to help bridge the gap between users' existing knowledge and their evolving information needs in the process of information foraging, enhancing the breadth and depth of their exploration while improving efficiency and reducing cognitive load.

The contributions of this paper are two-folds. First, we introduce a novel tool running as a web browser plug-in for displaying and managing AI-interpreted human intents in online information foraging tasks. Second, we contribute empirical findings from a preliminary user study with five users identifying the promises and challenges of this approach. Our findings suggested IntentPrism was a promising approach in facilitating web information foraging by explicitly visualizing and unifying LLM-extracted search intents in real time. The dynamic intent analysis and intent-based content highlighting has the potential to enhance users' experience in information foraging tasks by helping refine vague intents, facilitating easier expression and alignment of human and AI intentions, and improving focus on relevant information.

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¹<https://www.perplexity.ai/>

²<https://www.elmo.chat/>

2 Related Work

Information Foraging theory applies animal foraging principles to explain how humans search for information. It assumes users balance the expected value of information against its acquisition cost, leading to an optimal search strategy [18]. Specifically, It is crucial to help users efficiently locate relevant information, obtain intent-matching content, and avoid information overload in the era of information explosion on the internet. Although LLM-based conversational systems can provide quick answers to users' questions through summaries of relevant online information, Shah contended that the process of querying and discovery holds significant value, as it allows users to explore, encounter challenges, and learn, therefore he emphasized that search systems should help users find and make sense of information rather than tools that purport to do it all for them [21]. To address this, researchers are actively investigating various approaches to enhance support for exploratory search. Ruotsalo et al. introduced interactive intent modelling for exploratory search, where iterative keyword suggestions, informed by user feedback, enable refinement of search processes and adaptation to evolving goals [20]. Similarly, ConceptEVA addresses customized summaries for users from various domains, employing interactive visualizations to identify relevant content and generate summaries. Studies show participants preferred ConceptEVA's human-in-the-loop summaries over manually created ones, demonstrating its effectiveness in aiding comprehension [29]. However, both approaches rely on predefined concepts, limiting flexibility in supporting more abstract or personalized intents. Meanwhile, Scim facilitates rapid, high-level skimming of papers by highlighting relevant content based on a fixed taxonomy of four categories: Objective, Method, Results, and Novelty [7]. Studies found Scim improved information retrieval speed compared to standard readers. Nonetheless, its fixed taxonomy restricts goal-driven or personalized skimming. These tools highlight the potential of interactive systems but underscore the need for greater adaptability to evolving user intents. DiscipLink supports interdisciplinary information seeking (IIS) by addressing knowledge fragmentation and terminology gaps [30]. Its key innovations include the generation of exploratory questions to facilitate collaborative exploration, a query expansion mechanism based on pseudo-answers to bridge terminology differences, and the use of contextualized embeddings and topic extraction to enhance information organization efficiency. Meanwhile, Mindalogue explored a non-linear interaction paradigm with LLM based on mind mapping [28]. Users can freely control their exploration path without being constrained by linear conversations. By leveraging a visual structure, the system reduces cognitive load and enhances content credibility. User testing shows a 30% reduction in operational steps for completing the same tasks.

The research scope of IUIs (Intent-based user interfaces) can be divided into five interdependent aspects, ranging from user-centred to AI-centred perspectives: intent formation, intent expression, intent alignment, intent recognition, and intent execution. With LLM-driven interfaces, users gain the flexibility to express open-ended and nuanced intentions using natural language, but this often leads to challenges like ambiguity, missing context, or vague references [4, 19, 27]. Solutions typically involve either manual supplementation, such as slot filling with clarifying questions

[19], or automatic supplementation, like context-aware computing that integrates relevant environmental data [4]. Memolet, for example, allows users to manually organize historical conversations into cards for greater control and transparency in how LLMs interpret user intents [26]. Recent IUI systems are increasingly supporting users in refining and aligning their intentions across various tasks. Common approaches include allowing users to edit, add, or query outputs and user profiles representing system understanding [8, 9, 16]. For instance, PaTAT lets users modify or refine coding rules learned by an LLM based on user-provided examples [9]. While most systems assume independent and equally prioritized intentions, more complex tasks involve interdependent priorities, as demonstrated by Memolet, where users group and manage historical memos to handle these dependencies [26]. Intent recognition is often framed as a classification task, mapping user inputs to predefined categories such as functions, APIs, or navigation actions [2, 11]. For example, LAVE enables users to edit videos by modifying script-like text descriptions of keyframes [25]. While some studies aim to develop generalizable intent taxonomies [2], these often lack the flexibility to handle diverse, dynamic user goals. Shah proposed a Human-In-The-Loop method for generating adaptive taxonomies, but it still relies on static intent classifications from comprehensive datasets [22]. More dynamic approaches, like MeetMate, translate user intentions into mathematical constraints for meeting scheduling [14], but scaling this to complex scenarios may require even more advanced constraint models.

Our system differs from existing mindmap-like interfaces for interacting with LLM-generated content by integrating real-time interaction with original website content on the internet through bookmarking read information and highlighting new information through the browser plug-in. By testing this, we explore the effects of explicitly displaying and allowing user interaction with system-interpreted human intent in an online information foraging task: how it could help users clarify their mind and quickly identify more relevant information.

3 IntentPrism System

3.1 Interaction System Design

This section presents IntentPrism's interaction design philosophy and intention processing methodology. Our approach is based on Bratman's foundational work on intentions and planning [3, 10], integrating it with modern AI-assisted knowledge work contexts. The framework adapts the *Belief-Desire-Intention* (BDI) model [3, 10] to web-based knowledge work environments. The BDI model is a cognitive framework that represents the agency through three key mental attitudes: *Beliefs* (knowledge of the environment), *Desires* (motivational objectives), and *Intentions* (committed plans of action), which enables systems to understand their context, form goals, and develop actionable plans [13]. IntentPrism maps web interactions to the BDI model: browsing behaviors form *Beliefs*, annotations express *Desires*, and explicit confirmations establish *Intentions*.

3.1.1 Theoretical Framework and Design Consideration. In current Web environments, knowledge workers frequently interact with multiple information sources, leading to increased cognitive load

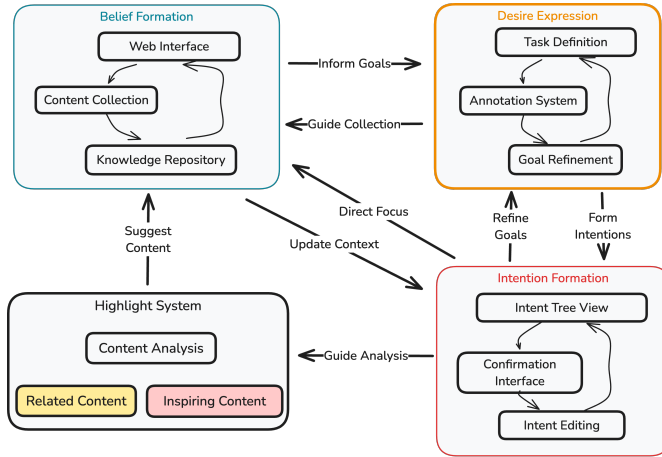


Figure 2: The IntentPrism system implements the BDI model through three interconnected modules. The *Belief* Formation module (teal blue) captures web content via interfaces and collection mechanisms into a knowledge repository. The *Desire* Expression module (orange) refines user goals through task definition, annotation, and goal refinement processes. The *Intention* Formation module (red) transforms the goal and implicit context into structured intentions via tree-based visualization and editing interfaces. The architecture features bidirectional information pathways, with the Highlight System analyzing and suggesting contextually relevant content. This system creates a cognitive cycle that adapts to evolving user needs while transforming information into actionable intentions.

and potential information loss. To mitigate these challenges, we mapped and extended the classic BDI model.

The architecture of the interaction system, depicted in Fig. 2, maps the web content collected through browsing and active annotations by users as *Beliefs*, which form the foundation of their current task knowledge and a shared background for communication between the user and the system. These *Beliefs* directly contribute to system functionality by providing the semantic foundation for the further intent recognition. *Desires* are divided into two levels representing a convergent process: on one side, the explicit task theme set by the user at the beginning; on the other side, the implicit needs expressed through specific annotations and comments during information gathering. These sources merge to shape the user’s *Desires* over time. *Intentions* are presented in a structured task decomposition tree that is built based on the user’s *Desires* and informed by their accumulated *Beliefs*. This combination generates concrete plans showing how users intend to accomplish their goals, presented in an intuitive visual structure that users can easily adjust throughout their information search process.

The BDI model was selected as the theoretical foundation for two primary reasons: first, it provides a clear cognitive framework for user decision-making; second, its layered structure facilitates the creation of transparent and explainable systems. Based on this, we introduced two key innovations: real-time belief construction

mechanisms that map user browsing behavior as a continuous accumulation process and dynamic *Intention* recognition mechanisms that maintain consistency in the *Intention* network while supporting dynamic changes.

3.1.2 Interaction Flow. The IntentPrism interaction system comprises four interrelated stages. First is the initial task definition phase, where users input their main task descriptions, establishing the foundational context as the root node for subsequent intention tree construction. Second, the information collection phase implements dual-mechanism content capture: a lightweight text selection menu with “Collect” and “Comment” buttons for rapid content acquisition and a accessible sidebar that serve as a centralized repository. This approach balances efficient content identification and annotation while maintaining uncluttered workspace organization.

Third, the *Intention* construction and visualization phase represents our most significant innovation. The system generates a three-level hierarchical visualization where the root node represents the main task, intermediate nodes reflect derived user intents automatically inferred through semantic relationship analysis, and leaf nodes contain collected content and annotations. Users maintain complete control through direct manipulation, with the ability to edit, add, or remove nodes to align with their cognitive models. To handle partial regeneration, we added a “Confirm” feature that allows users to lock validated parts of the intent tree.

Fourth, the intelligent content highlighting stage implements a dual-marker visual system, using yellow highlights to indicate content semantically similar to previously collected information and red highlights to identify novel but relevant content that may expand the user’s knowledge base. When viewing highlighted content, interactive hover functionality displays related intention nodes, establishing direct visual connections between current content and broader task contexts. This mechanism enables rapid relevance assessment while maintaining awareness of the overall task structure throughout the continuous knowledge work process.

3.2 Implementation

3.2.1 Dynamic Intent Analysis and Visualization. To achieve dynamic intent recognition, we draw inspiration from the LDA algorithm [12] and use GPT’s text analysis and reasoning capabilities to infer intents from user-selected text records. GPT first groups the records based on the user-defined Research Scenario, aiming to minimize intra-group and maximize inter-group differences. Then, GPT extracts a unique verb phrase from each group as its intent, ensuring logical distinction and avoiding overlap. Each intent is concise, clear, and limited to seven words. This process creates an intent tree, with Records as leaf nodes, Intents as intermediate nodes, and the Scenario as the root. Detailed prompts can be found in Appendix B.1 and B.2.

3.2.2 Intent-based Content Highlighting. The intent-based content highlighting process resembles Retrieval-Augmented Generation (RAG) but omits the final content generation step.

During highlighting, the system splits webpage content into sentences, uses each intent as a semantic query, and filters content based on relevance. We extended the intent by adding a description

to each node, following the idea of Query Expansion [1], to mitigate the inaccuracy of overly brief queries. In the highlighting phase, combining the intent with its description improves accuracy while keeping the front end concise.

Following the work by Fok et al., we first ask GPT to categorize sentences under appropriate intents and compute cosine similarity with the collected records [7]. If the similarity is below 0.25, the sentence is considered new content matching the intent. However, due to high cosine similarities (often around 0.7) without fine-tuning, distinguishing new information became challenging. We then tried using GPT to highlight text directly, which increased token usage and system latency. To address this, we added an overview of subtopics to the intent description and stopped sending chunks of records to GPT. Detailed prompts are provided in Appendix B.3.

4 User Study

We conducted a preliminary user study to evaluate the potential of this approach. Our research questions are: 1) How does explicit intent alignment between human and AI via our interface affect users' behavior and experience in information foraging tasks? 2) What are the promises and challenges of such intent alignment interfaces? 3) How could the system be improved to realize its potential?

4.1 Participants

We recruited 5 participants (aged 18-30, 3 female and 2 male) for an initial test of our prototype. The primary purpose of this test is to gather participant feedback and identify the prototype's strengths and weaknesses, which will inform our next iteration. We targeted participants who possess fundamental browser skills and have an interest in online information foraging or data collection. Three of them demonstrated a moderate to advanced level of familiarity with LLMs. We provided appropriate compensation (100 RMB per hour or an equivalent voucher) upon completion.

4.2 Procedure

Prior to the formal experiment, we briefed potential participants on the study's objectives, procedures, and possible risks and benefits, and provided them with a demonstration video as operational guidelines. After familiarizing themselves with the system, participants used our prototype to forage information on two topics of their choosing. Specifically, one topic focused on an in-depth exploration task, which involves conducting detailed research on a single theme to obtain comprehensive information and deep understanding. The other topic centred on a broad expansion task, aimed at establishing extensive connections across multiple related domains or themes, exploring diverse perspectives, and covering a wider range of content. Table.1 shows the detailed tasks that each participant conducted.

Throughout this process, we tracked front-end logs and observed participant actions, paying particular attention to how participants evolve their intents by interacting with the prototype, as well as any difficulties in understanding or using the interface. Once participants completed the task, we conducted a post-task interview and invited them to fill out a survey that included the System Usability Scale (SUS), along with subjective satisfaction ratings of key

features and open-ended questions about the system's functional effectiveness, usability, and potential improvements. This approach provided a comprehensive assessment of the system's usability. The experiment was conducted remotely via VooV Meeting, allowing us to observe and record participants' operations in real time. Qualitative data such as interview content will be subjected to thematic analysis.

5 Results

According to the post-task survey, participants gave a positive SUS rating for IntentPrism (Mean = 61.5, SD = 9.1). Feedback on functionality was also positive, with most users satisfied with intent analysis (Mean=3.8/5) and highlighting (Mean=3.8/5). The participants' system usage logs are visualized as activity timelines for every task as shown in Appendix A. We can observe their active usage of system functions throughout the task for aligning intention with AI, by recording content snippets, generating and editing the intent tree, and generating new highlights. One task for P2 was missing due to incompleteness caused by technical issues. We also conducted thematic analysis on the qualitative data collected from think-aloud sessions and post-task interviews. The process involved open coding to identify recurring patterns and key themes, followed by iterative refinement through axial coding to group related concepts. Through this approach, we obtained following key insights.

5.1 Benefits of the Approach

Supporting Intent Formation. Dynamic Intent Analysis enabled users to refine vague or coarse-grained intents into more specific and structured ones, reducing cognitive overload and enhancing efficiency in information foraging. As one participant (P3) mentioned, *"It helps me quickly build a mental framework to sort out the key points I need to focus on,"* which facilitated narrowing down priorities and reusing intent structures across webpages, minimizing redundant efforts. Another participant (P4) highlighted how the intent tree helped organize and refine broad categories into detailed subcategories, streamlining their thought process and enabling deeper exploration, *"It helps refine my somewhat vague ideas and make them more specific."* This ability to structure and refine intents improved overall efficiency.

Reduce Cognitive Load of Intent Management. Visualization aided users in memorizing and managing intents, especially for complex tasks. One participant (P3) noted, *"Since there's a lot of information, my mental framework keeps changing. For example, when I move to the next website, I may forget the intent structure I had for the previous one...I think the visualization helps me to memorize and manage all my intents."*

Assisting Intent Expression and Alignment. Underlining content and visualization streamlined the processes of intent expression and alignment, helping users clarify, refine, and adjust their intents, even when the system's initial analysis didn't fully match their expectations. One participant (P1) shared that *"the act of underlining itself makes me more aware of what my intentions actually are"*, transforming vague ideas into specific, concrete intents after analysing the content. Another participant (P2) highlighted the

Table 1: Tasks that participants chose

Participants	In-depth Exploration Task	Broad Expansion Task
P1	Learn How to Find a Job in a Specific Country and Create a Plan	Investigate the policy support for job searching in different countries.
P2	A One-Week Travel Guide to Japan	A Literature Review on LLM Teaching Assistant Systems (<i>Uncompleted due to technical issue</i>)
P3	Compare PLM Solutions	Conduct research on the topic of smart manufacturing.
P4	Compare the differences between P-Bass and J-Bass.	Summarize an overview of various Bass brands.
P5	Explore and compare the strengths and limitations of methods for named entity recognition of person names in sentences.	Conduct a methodological study on identifying users' idols based on platform user behaviour.

importance of making intent modification easier, as the system's analysis may not always align with their needs, and found the visualization format particularly helpful, stating *"In this regard, the visualization format indeed provides helpful support (for modifying intents)."* This approach enabled a more flexible and user-driven refinement of intents.

Enhancing Focus and In-depth Discovery. Intent-based content highlighting helped users efficiently identify and concentrate on relevant information, streamlining the information foraging process by focusing on highlighted content and uncovering overlooked yet important potential themes. One participant (P3) mentioned that highlighting allowed them to quickly skim through webpages and focus on relevant sections, *"For example, I noticed that the system highlighted a section titled 'Policy (of Smart Manufacturing)', which I hadn't considered before but turned out to be highly relevant to my topic."* This differentiation helped them identify new themes and enhance their research. The ability to filter content based on intent-specific highlights implies the potential to improve the efficiency and depth of information discovery.

5.2 In-depth Exploration vs. Broad Expansion

The participants' activity timelines shown in Appendix A reveals a potential trend that users utilized the "IntentTree" feature more frequently in Broad Expansion tasks, while highlighting was used more in In-depth Exploration tasks. While future research is needed to investigate this, we speculate that broad expansion tasks might rely more on the intent tree to support intent formation, whereas in-depth tasks benefit from quickly converging on an intent tree and using it extensively for highlighting.

Furthermore, participants found IntentPrism better suited for in-depth exploration tasks due to its effective refinement of user intents. One participant (P3) claimed that *"...because it involves understanding a relatively small and well-defined scope...and iterate on existing intents helps me quickly build a structured thought process and conduct searches efficiently."* Meanwhile, another participant (P2) also noted that the limited ability of IntentPrism to expand parallel or unrecognized intents makes it less suitable for broad expansion tasks, *"For instance, if I initially select Tokyo, Yokohama, and Mount Fuji as travel destinations...the system won't help me*

discover the additional possibilities (in Kyoto and Osaka)." Despite these limitations, one participant (P5) suggested that the system could help prevent missed aspects and support broad expansion tasks by proposing parallel intents.

5.3 Suggestions for System Improvement

Users expressed a preference for multi-layered intent structures, particularly in more complex tasks, as they help organize and plan intentions more effectively. For example, one participant (P2) noted, *"For travel, I prefer a multi-layered structure for the nodes. For example, I might use cities as primary nodes, and under each city, sub-nodes could represent activities like shopping or visiting historical sites."* Meanwhile, one participant (P5) also highlighted the need for a feedback mechanism to correct wrong highlights, *"The overall accuracy of the model in highlighting still needs improvement... I would like a way to provide feedback on incorrect highlights to help improve the system's performance in future interactions."* However, this raised a concern about scalability issue.

6 Limitation

This work is preliminary thus has the following limitations. First, the scalability of the visualization can be further improved. Three layers of the intent structure is a simple first step for testing this approach. Future work will extend this with more sophisticated design by following suggestions like Suh's work[24]. Second, we will conduct formal studies with more participants to understand more in depth users' interaction strategies with the intent expression and referred original web content.

7 Conclusion

This study shows that dynamic intent analysis and intent-based content highlighting has the potential to enhance users' experience in information foraging tasks. Participants reported that the system helped refine vague intents, facilitated easier expression and alignment of intentions, and improved focus on relevant information. While the system suits for assisting in-depth exploration tasks, it faces challenges in supporting broad expansion tasks due to its limited ability to expand parallel or unrecognized intents. The feedback suggests that a multi-layered intent structure would improve organization and support for more complex tasks, and that adding

a feedback mechanism for correcting inaccurate highlights could further enhance system performance. Overall, IntentPrism implies a promising design approach in facilitating web information foraging via explicit intent alignment. Future work will improve its flexibility and scalability of intent expression and conduct a more comprehensive evaluation. Broad information expansion tasks could be potentially supported by additional solutions such as topical suggestions.

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A Participant Activity Timeline

B Detailed Prompts

B.1 Prompt for Grouping Records in Dynamic Intent Analysis



Figure 3: Participant Activity Timeline of In-depth Exploration Tasks (I) and Broad Expansion Tasks (B). The markers represent different types of operations: orange cross for Task operations (creating new tasks); blue square for Record operations (saving selected text and adding comments); red circle for IntentTree operations (generating intent tree, deleting/editing/adding intent nodes); and purple triangle for Highlight operations (highlighting webpage content). The markers' colors are consistent with the contour colors of their corresponding function areas in Fig. 1.

```

1 You are a professional research assistant
  . Based on the given scenario , analyze
  the user's potential intent and use
  it as a basis to group the nodes
  provided in the list. When grouping ,
  ensure that inter-group differences
  are maximized while intra-group
  differences are minimized. Return a
  list of sublists , where each sublist
  represents a group, and the elements
  of the sublist are the nodes from the
  original list .
2
3 # Output Format
4   - The output should be structured in
    JSON format as following {
      format_instructions }.
5
6 # User:
7   Scenario: {scenario}
8   List: {list}

```

B.2 Prompt for Analysing Intents in Dynamic Intent Analysis

```

1 You will act as an assistant to help the
  user conduct research based on the
  given Scenario. Please extract a
  research intent for each dictionary
  element in Groups and return the
  results in dictionary format. Ensure
  that each intent is clearly described ,
  non-repetitive , and consistent in
  granularity .
2
3 # Steps
4   1. **Confirm the number of groups**:
     Understand the structure of Groups
     and confirm the number of groups.
     The number of extracted intents
     should match the number of groups.
     A group is defined as a
     dictionary with group_x as the key
     and a list as the value .

```

```

5      2. **Understand the scenario**:
        Analyze the meaning of the
        Scenario and infer the user's
        potential intent in this context.
6      3. **Extract intents**: For each
        group, extract a corresponding
        intent. The intent should align
        with the commonalities among all
        Nodes in the group while
        incorporating user comments to
        refine and ensure relevance. Each
        intent must be closely related to
        the Scenario, logically distinct,
        and non-overlapping. Each intent
        should be a concise verb phrase of
        no more than 7 words. The
        description should clearly explain
        the theme behind the intent and
        the subtopics already covered in
        the group.
7      4. **Create a new dictionary**:
        Following the Output Format, use
        the extracted intents as keys and
        the corresponding group dictionary
        keys as values.
8      5. **Compare and replace**:
9      - Compare the intents in
        IntentsList with the extracted
        intents in the dictionary to
        find the most similar intent.
10     - Replace the dictionary intent
        with the most similar one from
        IntentsList if a match is
        found.
11     - Retain the original intent if
        no sufficiently similar match
        is found.
12     6. **Add remaining intents**: For
        unused intents in IntentsList,
        treat them as remaining_intent.
        Add them to the new dictionary
        created in step 4 with their
        values set as empty strings. Skip
        this step if all intents are used.
13     7. **Add descriptions**: For every
        intent in the new dictionary,
        explain the reason why to
        construct the intent, and also
        provide a specific description of
        the subtopics covered in the group
        .
14
15 # Output Format
16 - The output should be structured in
        JSON format as following {
        format_instructions}.
17 - example:

```

```

18     {{ 'item': {{
        generated_intent_1': {{
            group": "group1", "
            description": "Reasons and
            theme for
            generated_intent_1.
            Existing sub-themes:
            sub_theme_1, sub_theme_2
            "}}, 'generated_intent_2 ':
            {{ "group": "group2", "
            description": "Reasons and
            theme for
            generated_intent_2.
            Existing sub-themes:
            sub_theme_3, sub_theme_4
            "}}, 'remaining_intent_3 ':
            {{ "group": "", "
            description": "Reasons and
            theme for
            generated_intent_1.
            Existing sub-themes
            : "}}}}}}
19     - 'generated_intent_x', 'sub_theme_x'
        and 'remaining_intent_x' should
        be replaced with specific intent
        phrases.
20
21 # Notes
22 - Extract exactly one unique intent
        per group. The intent must
        summarize the commonalities of all
        Nodes in the group.
23 - Each intent description must be
        logically independent and maintain
        diversity to the greatest extent,
        without exceeding 7 words.
24 - When performing similarity
        replacement, ensure that
        replacement is only made if
        similarity is sufficiently high.
25 - If no sufficiently similar intent
        exists, the original intent
        remains unchanged, and unused
        intents in IntentsList will not be
        forcibly replaced.
26 - Verify the final generated
        dictionary to ensure all intents
        are at the same level of
        granularity. Adjust intent texts
        if necessary.
27
28 # User:
29     Scenario: {scenario}
30     Groups: {groups}
31     IntentsList: {intentsList}

```

B.3 Prompt for Intent-based Content Highlighting

```

1  ## System
2  You are tasked with assisting the user in
   conducting information research based
   on a specified scenario. Your goal is
   to filter sentences from a given
   SentenceList for each Intent provided
   in the IntentsDict.

3
4  ### Steps
5  1. **Understand Input Structure**:
6     - **Scenario**: Describes the
       research context, providing high-
       level guidance..
7     - **SentenceList**: A list of
       sentences.
8     - **IntentsDict**: A dictionary
       structured as `{"intent": "
       description"}`
9     - Each key represents an intent,
       and the value (description)
       summarizes the theme and
       relevant sub-themes of the
       intent.

10
11  2. **Filter and Match Sentences**:
12     For each `Intent` and its
       corresponding `Description` in
       IntentsDict:
13     - **Description Understanding**:
14         - Understand the reason why user
           has the `Intent` and the
           collected sub-themes of `
           Intent` in `Description`.
15         - If `Description` is empty,
           analyze the reason of the `
           Intent` based on the `Scenario`
           `.
16     - **Sentence Evaluation**: For each
       sentence in `SentenceList`:
17         - If `Description` or the `
           Existing sub-themes` in `
           Description` is empty, skip
           the step `Align with Intent
           Reason and Sub-Themes`.
18         - Align with Intent Reason and
           Sub-Themes:
19             - If the sentence aligns with
               both the reason in `
               Description` and collected
               sub-themes of the intent,
               add it to top_all[Intent
               ].
20         - Align with Intent Reason Only:

```

```

21         - If the sentence aligns only
           with the reason in `
           Description` and not with
           sub-themes, add it to
           bottom_all[Intent].

22
23  3. **Relevance Check**:
24     - Avoid including sentences in the
       output unless the sentences
       genuinely align with the intent's
       theme or sub-themes.
25     - If no sentences match for a
       specific intent, return an empty
       list for top_all[Intent] or
       bottom_all[Intent] as applicable.
       If no sentences match the criteria
       , return an empty list for `
       top_all[Intent]` or `bottom_all[
       Intent]` as applicable.

26
27  ### Output Format
28     - The output should be a dictionary with
       two main keys: `top_all` and `
       bottom_all`.
29     - {format_instructions}
30     - `bottom_all`: Maps each Intent to a
       list of sentences corresponding
       to sentences aligned only with the
       reason.
31     - `top_all`: Maps each Intent to a
       list of sentences corresponding to
       sentences aligned with both the
       intent's theme and sub-themes.

32
33  ### Notes
34     - Relevance Over Quantity: Only
       include sentences that fully meet
       the alignment criteria. Avoid
       adding sentences simply to
       increase the count.
35     - Empty Description Handling: If the
       Description of Intent is empty,
       skip the step `Align with Intent
       Reason and Sub-Themes`.
36     - Top_all vs. Bottom_all: Maintain a
       clear distinction between
       sentences qualifying for top_all (
       aligned with both the theme and
       sub-themes) and bottom_all (
       aligned only with the theme).
37     - Exhaustive Intent Coverage: Ensure
       every intent in IntentsDict is
       represented in the output, even if
       no sentences align with it.

```

```
38 | - Avoid Duplication: Prevent any
    | sentence from appearing multiple
    | times across or within top_all and
39 | bottom_all lists.
```

```
40 | ## User:
41 |     Scenario: {scenario}
42 |     IntentsDict: {intentsDict}
43 |     SentenceList: {sentenceList}
```